

Ervin Samuel M.Sc.

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ML engineer with M.Sc. in Computing Science, experienced in end-to-end ML pipelines, model training and deployment, and integrating AI into production — with a 95% performance improvement track record at scale.

MACHINE LEARNING RESEARCH

Simon Fraser University, Autonomous Intelligence and Robotics Lab

Sep 2022 – Dec 2025

Learning-Based Optimization for Large-Scale Multi-Agent Planning (supervised by Prof. Hang Ma)

- Designed and deployed an end-to-end ML automation pipeline — ingesting raw solver traces, training a priority-prediction model, and integrating it into a live C++ planning system — reducing planning latency by up to 95% while maintaining solution quality within 6%.
- Built dataset curation logic to correct a ~99% class imbalance via targeted undersampling of rare conflict pairs, enforcing data quality standards that ensured reliable model behavior in production.
- Implemented a CNN + multi-head attention model in PyTorch encoding agent observations into learned spatial and coordinate embeddings, cutting search tree expansions by ~97%.
- Developed confidence thresholding and fail-focused edge removal so the system degrades gracefully when predictions over-constrain the planner, preserving reliability under distribution shift.
- Ran ablation studies across confidence thresholds and edge-removal strategies to validate each system component's contribution and establish reliable evaluation metrics.

PROJECTS

Job Match Intelligence System (job-match-intelligence.streamlit.app)

May 2026

- Built an end-to-end RAG pipeline that ingests job postings from Gmail, ranks them by semantic similarity to a candidate profile, and generates structured fit analysis using GPT-4o.
- Full stack: Gmail OAuth ingestion, OpenAI embeddings (1536-dim), ChromaDB vector store, GPT-4o reasoning layer, and Streamlit UI deployed to Streamlit Cloud.
- Implemented an LLM-as-judge evaluation framework measuring relevance, faithfulness, and actionability of generated reasoning across 157 job postings.

ENGINEERING EXPERIENCE

Neurelli

Feb 2022 – May 2022

Machine Learning Engineer Intern

Linkou, Taiwan

- Built an end-to-end ML automation pipeline — ingestion, cleaning, data governance, feature engineering, and XGBoost model training — to predict logistics delivery delays from structured operational data.
- Monitored model outputs against production outcomes to detect and correct feature drift, iterating on feature design and revalidating on updated data to maintain reliability.

HTC Corporation

Jun 2021 – Aug 2021

Software Engineer Intern

Taipei, Taiwan

- Built REST API validation tooling using Robot Framework, automating integration testing across backend services and reducing manual QA effort for a VR platform shipped to millions of users.

TECHNICAL SKILLS

Languages:	Python, C++
Machine Learning:	PyTorch, XGBoost, Scikit-learn, Supervised Learning, RAG
ML Systems:	Data Pipelines, Feature Engineering, Data Validation, Real-Time Inference, C++/Python Integration, REST APIs
Tools & Data:	NumPy, Pandas, Git, Linux, Jupyter, TensorBoard, OpenAI API, ChromaDB, Streamlit

EDUCATION

Simon Fraser University

Sep 2022 – Dec 2025

Master of Science in Computing Science, GPA: 3.84/4.33

Burnaby, Canada

Thesis: "Learning to Prioritize: Accelerating Priority-Based Search for Lifelong Multi-Agent Path Finding."

National Tsing Hua University

Sep 2018 – Jun 2022

Bachelor of Science in Computer Science, GPA: 3.94/4.3

Hsinchu, Taiwan

PART TIME EXPERIENCE

Simon Fraser University

May 2023 – Dec 2025

Teaching Assistant

Burnaby, Canada

- Supported 100+ students in AI and Data Structures & Algorithms through office hours and grading.